Spatial Simpson integration for reflected-clock iterated Brownian motion

Project Stardust

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Abstract

Let $Z_t = W(|B_t|)$, where W and B are independent standard Brownian motions. We prove convergence in probability of oriented spatial Simpson sums for every global C^3 one-form. The limit is its ordinary spatial primitive evaluated along Z. The proof uses a compact fourth-order Peano remainder and a direct quartic tightness estimate, without the sharper quartic-variation limit. We then prove a deterministic nonseparable C^1-time/C^4-space chain formula, uniform control over prefixes of one common grid, bounded random-horizon convergence for spatial one-forms, and an explicit Bernstein realization converging in continuous-path supremum norm. The time and space hypotheses, measurability, and limiting topologies are stated separately. These prescribed symmetric quadratures are not identified with a causal physical-time divergence. The principal results have exact declarations in a frozen Lean development; the exposition supplies full human proofs. Novelty of this package is unresolved.

1 The process and the quadrature prescription

Work on a probability space carrying independent real standard Brownian motions $B = (B_t)_{t\geq 0}$ and $W = (W_s)_{s\geq 0}$, starting at zero, with continuous versions. Define

$$Z_t = W(|B_t|), \qquad t \geq 0. \tag{1}$$

Thus the clock is reflected and the scenery is indexed by nonnegative spatial time. This convention is fixed throughout. The product Wiener realization is equivalent for these statements to the constructed independent Gaussian scenery–clock realization used in the formal sources. The process is jointly measurable and has continuous paths on a common full-measure set. All samplewise arguments below are on that set.

For a real function g, define the oriented spatial Simpson increment

$$S_g(x,y) = \frac{y-x}{6}\left(g(x) + 4g\left(\frac{x+y}{2}\right) + g(y)\right). \tag{2}$$

The sign of $y - x$ is retained. The midpoint is the spatial midpoint of the two endpoints; it is not evaluation of Z at the midpoint of a physical time interval.

Fix $T > 0$, use all uniform grids $t_i = iT/N$, $0 \leq i \leq N$, and abbreviate $Z_i = Z_{t_i}$. Define

$$P_{g,N}(k) = \sum_{i=0}^{k-1} S_g(Z_i, Z_{i+1}), \qquad 0 \leq k \leq N.$$

The finite sums are defined directly from the integrand and the process values. The limiting primitive will be proved to agree with them, rather than being substituted as their definition.

Theorem 1.1. *Let $g \in C^3(\mathbb{R})$ and set*

$$J_g(t) = \int_{Z_0}^{Z_t} g(x)\,dx,$$

where the integral is oriented. Then J_g has continuous paths, is jointly measurable, and

$$\max_{0\le k\le N} |P_{g,N}(k) - J_g(t_k)| \xrightarrow{\mathrm{P}} 0. \tag{3}$$

No global growth bound on g or its derivatives is assumed.

The uniformity in (3) is over prefixes of one common grid. It is not an assertion that arbitrary partitions or arbitrary adapted time-dependent integrands are covered. The deterministic time–space formula and the continuous-path realization will have their own definitions below.

Stochastic integration for iterated Brownian motion has classical antecedents. Khoshnevisan and Lewis [1] use a two-sided scenery and a random partition induced by a spatial clock grid. Our reflected-clock convention and deterministic physical-time grids are different. Simpson-type stochastic sums have also been studied for fractional Brownian motion [2]; covariance or scaling similarity does not identify those processes with (1). The paper gives a verified, domain-explicit exposition, not a claim of first construction of symmetric integration for iterated Brownian motion.

2 A direct quartic tightness estimate

For an interval $[a, b]$ let $\Delta Z = Z_b - Z_a$. Conditional on the complete clock path, the two scenery times are fixed and ΔZ is a centered Gaussian variable of variance

$$L = \big||B_b| - |B_a|\big|.$$

Independence of W and B is essential here. Reversal of the order of the scenery times only changes the sign of a Brownian increment and not its Gaussian fourth moment. Therefore

$$\mathbb{E}[(\Delta Z)^4 \mid B] = 3L^2, \qquad \mathbb{E}[(\Delta Z)^4] \le 3(b-a), \tag{4}$$

using $L \le |B_b - B_a|$ and the Brownian increment variance. These expectations can equally be computed by Fubini on the independent scenery–clock product space, so no unspecified conditional version is required.

Put

$$Q_{4,N} = \sum_{i=0}^{N-1} |Z_{i+1} - Z_i|^4, \qquad M_N = \max_{i<N} |Z_{i+1} - Z_i|.$$

Equation (4) yields

$$\mathbb{E}Q_{4,N} \le 3T, \qquad \sup_N \mathbb{P}\{Q_{4,N} > A\} \le \frac{3T}{A} \quad (A > 0). \tag{5}$$

Thus the quartic sums are tight. No independence between different $\Delta_i Z$ is needed when summing their expectations. Continuity of each path on $[0, T]$ gives $M_N \to 0$ almost surely.

The sharper assertion $Q_{4,N} \xrightarrow{\mathrm{P}} 3T$ is separately verified in the library, but is not needed for any main proof in this paper. In particular it would be incorrect to infer that assertion from (5) or the corresponding expectation calculation. The present argument uses only the bound it has actually proved.

Lemma 2.1. *If $U_N \to 0$ in probability and $V_N \geq 0$ is tight, then $U_N V_N \to 0$ in probability.*

Proof. For $A > 0$ and $\varepsilon > 0$,

$$\mathbb{P}\{|U_N V_N| > \varepsilon\} \leq \mathbb{P}\{V_N > A\} + \mathbb{P}\{|U_N| > \varepsilon/A\}.$$

First choose A to make the first term uniformly small, then let $N \to \infty$ in the second. □

This elementary transfer principle is why a local remainder modulus is sufficient; neither an almost-sure variation bound nor global bounds on the integrand are needed.

3 Compact Simpson remainders

We now establish the deterministic estimate which controls every stochastic prefix simultaneously.

Lemma 3.1. *Let $g \in C^3(\mathbb{R})$ and $R > 0$. There is a nonnegative modulus $\omega_R(\delta) \to 0$ as $\delta \downarrow 0$ such that, for $|x|, |y| \leq R$,*

$$\left| S_g(x, y) - \int_x^y g(z)\, dz \right| \leq \frac{1}{3}\omega_R(|y - x|)|y - x|^4. \tag{6}$$

One may take ω_R to be the modulus of continuity of g''' on $[-R, R]$.

Proof. Let $d = y - x$. The case $d = 0$ is immediate. On the segment between x and y, Taylor's theorem in integral form gives

$$g(x + \theta d) = g(x) + g'(x)\theta d + \frac{g''(x)}{2}\theta^2 d^2 + \frac{g'''(x)}{6}\theta^3 d^3 + r(\theta)$$

for $0 \leq \theta \leq 1$, with $|r(\theta)| \leq \omega_R(|d|)|d|^3/6$. Indeed, subtract the constant $g'''(x)$ inside the third-order integral remainder and use its uniform continuity. This proof works for either sign of d.

The Simpson functional on $[0, 1]$, with weights $1/6, 4/6, 1/6$, integrates polynomials of degree at most three exactly. Directly, its values on $1, \theta, \theta^2, \theta^3$ are $1, 1/2, 1/3, 1/4$. Thus all displayed polynomial terms cancel against the actual integral. The Simpson average of the remainder and its integral each have absolute value at most $\omega_R(|d|)|d|^3/6$. Multiplying their difference by $|d|$ proves (6). □

The estimate is fourth order with a vanishing compact coefficient. It requires g''' continuous, not a fourth derivative of g. Equivalently, when $g = f'$, it requires a C^4 primitive, not a fifth derivative. This is the regularity boundary used in the formal compact Peano result.

For a continuous path $z : [0, T] \to \mathbb{R}$ with range in $[-R, R]$, sum (6) over any grid prefix. Since the exact oriented integrals telescope through the ordinary primitive,

$$\max_{k \leq N} \left| \sum_{i<k} S_g(z_i, z_{i+1}) - \int_{z_0}^{z_k} g \right| \leq \frac{1}{3}\omega_R(M_N(z))Q_{4,N}(z). \tag{7}$$

There is one modulus and one full nonnegative quartic sum for every prefix. No union bound with a factor growing in N is being applied.

4 The spatial formula and measurable random horizons

Proof of Theorem 1.1. The function $G(x) = \int_0^x g(z)\,dz$ is C^4 by the fundamental theorem of calculus, so $J_g(t) = G(Z_t) - G(Z_0)$ is continuous in t and jointly measurable. Each finite Simpson sum is measurable because it is a finite sum of continuous functions of measurable coordinates.

The range of a continuous sample path is bounded, so for any $\eta > 0$ choose deterministic R with $\mathbb{P}\{\sup_{t\le T} |Z_t| > R\} < \eta$. On its complementary event (7) applies. There $\omega_R(M_N) \to 0$ almost surely. Quartic tightness (5) and Lemma 2.1 show that the right side tends to zero in probability. The probability of the excluded event is less than η, which is arbitrary. This proves the uniform-prefix assertion. □

Corollary 4.1. *Let τ be any measurable random time with $0 \le \tau \le T$. Define*

$$k_N = \left\lfloor \frac{N\tau}{T} \right\rfloor, \qquad \tau_N = \frac{T}{N} k_N.$$

Then $P_{g,N}(k_N) \xrightarrow{\mathrm{P}} J_g(\tau)$ for $g \in C^3(\mathbb{R})$.

Proof. The integer index is measurable, takes values in $\{0, \dots, N\}$, and obeys $0 \le \tau - \tau_N < T/N$. Evaluating the finite collection of measurable prefix sums at that index is measurable. The error to $J_g(\tau_N)$ is bounded by the prefix maximum in (3). Pathwise continuity gives $J_g(\tau_N) \to J_g(\tau)$ almost surely, proving the result. □

No stopping-time hypothesis is used. For example τ can equal T or $T/2$ according to the sign of Z_T. This conclusion uses uniform approximation and continuity, not optional sampling of a martingale.

5 Deterministic time–space integrands

Let $F : \mathbb{R}^2 \to \mathbb{R}$ satisfy the following hypotheses: for every fixed t, $F(t, \cdot) \in C^4(\mathbb{R})$; for every fixed x, $F(\cdot, x) \in C^1(\mathbb{R})$; and the functions $\partial_t F$ and $\partial_x^4 F$ are jointly continuous. No mixed derivative or global derivative bound is assumed. For the continuous-path realization in Section 7, we additionally state joint continuity of F explicitly, as in its formal declaration.

Define the time-frozen spatial sums

$$P_{F,N}(k) = \sum_{i<k} S_{\partial_x F(t_i,\cdot)}(Z_i, Z_{i+1}), \tag{8}$$

and the continuous target

$$J_F(t) = F(t, Z_t) - F(0, Z_0) - \int_0^t \partial_t F(s, Z_s)\,ds. \tag{9}$$

The target is continuous. In fact the stated hypotheses already give joint continuity of F: near a fixed time t_0, write $F(t, x) = F(t_0, x) + \int_{t_0}^t \partial_t F(s, x)\,ds$ and use continuity and local boundedness of the joint time partial. We keep the explicit hypothesis for the later source correspondence.

Theorem 5.1. *Under these time–space hypotheses,*

$$\max_{k\le N} |P_{F,N}(k) - J_F(t_k)| \xrightarrow{\mathrm{P}} 0. \tag{10}$$

In particular the full sums converge in probability to (9) *at $t = T$.*

Proof. On a compact rectangle $[0,T]\times[-R,R]$, joint continuity of $\partial_x^4 F$ gives one horizontal modulus uniform in t. Apply Lemma 3.1 with $g=\partial_x F(t,\cdot)$. The proof uses only differences of $\partial_x^4 F$ at nearby spatial points, so its modulus works for every frozen time. Thus the maximum error in replacing (8) by

$$V_{F,N}(k)=\sum_{i<k}\big(F(t_i,Z_{i+1})-F(t_i,Z_i)\big)$$

tends to zero in probability, by exactly the same localization and quartic-tightness argument as before.

It remains to identify $V_{F,N}$. The exact finite chain identity is

$$\begin{aligned}F(t_k,Z_k)-F(0,Z_0)&=V_{F,N}(k)\\&\quad+\sum_{i<k}\big(F(t_{i+1},Z_{i+1})-F(t_i,Z_{i+1})\big).\end{aligned}\tag{11}$$

The time fundamental theorem of calculus identifies the second sum as

$$\sum_{i<k}\int_{t_i}^{t_{i+1}}\partial_t F(s,Z_{i+1})\,ds.$$

For each fixed continuous sample path, uniform continuity of $\partial_t F$ on its compact time–space range shows that its difference from $\int_0^{t_k}\partial_t F(s,Z_s)\,ds$ is bounded, uniformly over k, by

$$T\,\nu_R\!\left(\sup_{|t-s|\le T/N}|Z_t-Z_s|\right)\longrightarrow 0.$$

Here ν_R is a horizontal modulus of the time partial on the compact rectangle. This is a pathwise limit. Combining it with the spatial error and (11) proves (10). □

The time correction is an ordinary finite Lebesgue integral on each sample path. It is not an assumed stochastic correction term. Joint measurability of $\partial_t F(s,Z_s)$ and integration over a bounded interval make it a measurable random variable. The same argument gives measurability of the finite sums, since each frozen spatial slice has a continuous derivative.

6 Weights and elementary checks on the prescription

Taking $F(t,x)=h(t)G(x)$, where $h\in C^1(\mathbb{R})$ and $G'=g\in C^3(\mathbb{R})$, gives

$$\sum_{i<N}h(t_i)S_g(Z_i,Z_{i+1})\xrightarrow{\mathrm{P}}h(T)J_g(T)-\int_0^T h'(s)J_g(s)\,ds.\tag{12}$$

The equality of this target with the one in Theorem 5.1 follows from cancellation of the constant $G(Z_0)$ and $\int_0^T h'=h(T)-h(0)$. Thus the separable time-weighted rule is contained in the full nonseparable theorem. Random time coefficients are not included by replacing h with a random function without further work.

The spatial rule is exact for cubic one-forms. For example

$$S_{4z^3}(x,y)=y^4-x^4.$$

It is not always an exact increment of a primitive: direct polynomial expansion gives

$$S_{5z^4}(x,y)-(y^5-x^5)=\frac{(y-x)^5}{24}.\tag{13}$$

For a continuous path the sum of absolute fifth powers is at most $M_N Q_{4,N}$, which tends to zero in probability here. This example illustrates the finite prescription and its limiting primitive without conflating them.

The sign in (2) also matters. Replacing $y - x$ by $|y - x|$ would already destroy the oriented integral for a constant one-form. Similarly, replacing the spatial midpoint by $Z_{(t_i+t_{i+1})/2}$ changes the approximation scheme; the deterministic Simpson polynomial cancellations no longer apply to that expression.

7 An explicit continuous-path realization

Under the time–space hypotheses, including explicit joint continuity of F, define on $0 \le u \le 1$

$$A_{F,N}(u) = \sum_{k=0}^{N} \binom{N}{k} u^k (1-u)^{N-k} P_{F,N}(k). \tag{14}$$

This is an actual random polynomial assembled from the prefixes of the one common grid. Let $j_F(u) = J_F(Tu)$.

Theorem 7.1. *As random elements of $C([0,1],\mathbb{R})$ with supremum norm,*

$$\|A_{F,N} - j_F\|_\infty \xrightarrow{\mathrm{P}} 0. \tag{15}$$

The approximants and target are Borel measurable in this continuous-path space.

Proof. The Bernstein basis is nonnegative and sums to one, so

$$\|A_{F,N} - B_N j_F\|_\infty \le \max_{k \le N} |P_{F,N}(k) - j_F(k/N)|, \tag{16}$$

where B_N is the ordinary Bernstein approximation operator. The right side tends to zero in probability by Theorem 5.1.

For a fixed continuous f, Bernstein approximation gives $\|B_N f - f\|_\infty \to 0$. A direct proof writes $(B_N f)(u) = \mathbb{E}[f(K/N)]$ for a binomial K of parameters N, u. Split at $|K/N - u| \le \delta$ and use $\mathbb{E}(K/N - u)^2 = u(1-u)/N \le 1/(4N)$ to obtain

$$\|B_N f - f\|_\infty \le \omega_f(\delta) + \frac{2\|f\|_\infty}{4N\delta^2}.$$

First take $N \to \infty$ and then $\delta \downarrow 0$. Apply this pathwise to the continuous target j_F. The resulting almost-sure Bernstein error tends to zero in probability. Combine it with (16).

Each coefficient in (14) is measurable, hence the finite polynomial is a measurable $C([0,1])$-valued map. For the target, rational evaluation maps are measurable by the measurability arguments following Theorem 5.1. The Borel sigma algebra of the separable space $C([0,1])$ is generated by these evaluations: suprema of continuous-path distances equal suprema over a countable dense subset. Thus the target path map is measurable as well. □

The approximation has $A_{F,N}(0) = 0$ and $A_{F,N}(1) = P_{F,N}(N)$, since the Bernstein basis selects the first and last coefficients at the endpoints. At intermediate times, all grid prefixes can enter (14). Consequently it is a polynomial smoothing, not a claimed causal interpolation. No adaptedness or L^p convergence follows merely from (15).

8 What this integration result means

The targets are genuine pathwise finite spatial or time integrals, and the approximations converge under the actual positive scenery–clock law. Here "oriented" refers to the signed spatial increment $y - x$, not to a signed probability measure for Z. This distinguishes the present setting from the higher-order signed heat pseudoprocess calculus.

The proof is local in the spatial range. It admits unbounded global C^3 one-forms and C^4 spatial slices because every continuous sample path has a compact range. It provides convergence in probability, not integrability of arbitrary unbounded targets. Neither expectation convergence nor strong L^2 convergence has been asserted without moment hypotheses.

It also does not construct a general physical-time adapted stochastic integral. Its spatial one-form depends only on the current spatial argument, and its time–space integrand comes from a deterministic primitive F. The rule has not been identified with the original intrinsic divergence or with an operator acting on arbitrary predictable integrands. Bounded random horizons are handled by approximation, not a martingale stopping theorem. Nonsmooth Itô–Tanaka formulas and arbitrary partitions have their own proof obligations.

Earlier library notes also discuss raw Taylor-channel divergence and intrinsic graph-dual obstructions. Those results are preserved, but they are not needed as premises of the proofs here. Keeping them separate avoids using a stochastic-domain assertion to justify a deterministic quadrature estimate.

9 Formal correspondence and preliminary comparisons

The frozen mathematical baseline is

`bab688bb6140cd6316c80680095a914ea0d346db.`

The namespace prefix is

`OperatorDerivativeRewrite.Foundation`

The companion `theorem-map.json` contains exact declaration links, source line numbers and hashes.

Paper result	*Lean namespace and declaration*
Compact spatial estimate	`IBMSimpsonPeano.compact_error_bound`
Spatial convergence	`IBMActualSimpson.sums_tendstoInMeasure`
Uniform spatial prefixes	`IBMUniformSimpsonProcess.maximum_tendstoInMeasure`
Random horizons	`IBMSimpsonRandomHorizon.stopped_sums_tendstoInMeasure`
Time–space compact estimate	`IBMTimeSpacePeano.compact_error_bound`
Time quadrature	`IBMTimeSpaceQuadrature.increments_tendsto`
Time–space chain	`IBMTimeSpaceSimpson.sums_tendstoInMeasure`
Uniform time–space prefixes	`IBMUniformTimeSpace.maximum_tendstoInMeasure`
Time weights	`IBMTimeWeightedSimpson.weighted_tendstoInMeasure`

The continuous-path statements are `IBMContinuousSimpsonRealization.approximation_error_bound` and `approximation_tendstoInMeasure` in the same namespace.

The formal proof uses its verified quartic-variation machinery to control compact Peano errors. The human proof above reorganizes the same application using the weaker direct Gaussian moment bound (5). That bound is proved explicitly here; no separate named Lean declaration for this shortened argument is claimed. The endpoints, smoothness assumptions and probability topologies of the principal convergence statements are those in the formal declarations. The non-separable chain theorem in the source does not require joint continuity of F as a separate input; the continuous-path declaration retains it explicitly.

The baseline receipt records a complete successful Lean build and 54,943 declarations audited with only the standard three axioms. This editorial work changes no mathematical source and does not claim a new full build. Source hashes, baseline evidence, compiler diagnostics and visual-review receipts accompany the paper. The earlier signed-Simpson working note remains intact; this paper replaces its mechanism summaries with proofs for the focused quadrature and continuous-path scope.

The primary comparison [1] constructs symmetric integrals for a two-sided-scenery IBM using an induced random partition. It is not an identification with the reflected-clock convention or deterministic grids used here. Harnett and Nualart [2] study the critical fractional Brownian Simpson problem and distributional limits. These are close antecedents for numerical prescriptions, but their process and limiting regime differ. The compact-remainder method itself is elementary; this paper does not turn its formal verification into a novelty claim. Priority for the combined domain, prefix and path-space package requires further targeted comparison.

References

[1] D. Khoshnevisan and T. M. Lewis, Stochastic calculus for Brownian motion on a Brownian fracture, *Ann. Appl. Probab.* **9** (1999), 629–667. `https://www.math.utah.edu/~davar/ps-pdf-files/ibm.pdf`.

[2] D. Harnett and D. Nualart, On Simpson's rule and fractional Brownian motion with $H = 1/10$, *J. Theor. Probab.* **28** (2015), 1651–1688. doi:10.1007/s10959-014-0552-1.